

Kashif Mahmood

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ABOUT ME

I'm Kashif Mahmood, a Software Engineering graduate specializing in artificial intelligence and data science, with research experience spanning medical imaging, agricultural AI, and applied analytics. My work includes building and publishing a self-supervised learning framework for chest X-ray classification, independently creating and documenting a public image dataset, and delivering measurable results in applied data analytics roles. I'm continually expanding my technical range across deep learning, MLOps, and reproducible research practice.

WORK EXPERIENCE

 **University of Sargodha** – Sargodha, Pakistan

Research Assistant

[09/2024 – Current]

- Served as lead or co-author on three research manuscripts spanning agricultural and medical imaging AI, including a self-supervised learning framework for chest X-ray classification and a transfer-learning approach to peanut disease detection, both at *Scientific Reports*.
- Independently conceived, built, and documented a public image dataset of 8,944 peanut images, including custom-built data-preparation software.
- Conducted literature reviews, experimental design, and model development across deep learning and self-supervised learning projects, in collaboration with international research partners at the Islamic University of Madinah and Princess Nourah bint Abdulrahman University.

 **10Pearls** – Remote, Pakistan

Data Science Intern (Part-time, concurrent with Research Assistantship)

[10/2025 – 11/2025]

- Built a fully automated, end-to-end 72-hour Air Quality Index (AQI) forecasting system for Karachi's NED University region, from raw API ingestion through deployment.
- Engineered the machine learning pipeline: feature engineering, model evaluation across four algorithms via GridSearchCV, and multi-step forecasting via a MultiOutputRegressor, with all features and models versioned in Hopsworks.
- Automated the entire pipeline with dual GitHub Actions workflows for hourly data ingestion and daily model retraining, requiring no manual intervention.

 **BrandPur** – Sargodha, Pakistan

Data Analytics Intern

[06/2024 – 08/2024]

- Analyzed customer data to understand retention and churn behavior, using Python (pandas, matplotlib, seaborn) for analysis and Power BI for dashboarding.
- Built a churn-prediction model estimating the probability of customer attrition, which informed targeted retention packages for at-risk customers.
- Contributed to an approximate 10% increase in customer retention through this data-driven retention strategy.

EDUCATION AND TRAINING

Bachelor of Science in Software Engineering

University of Sargodha, Pakistan [10/2021 – 06/2025]

City: Sargodha | Country: Pakistan | Website: <https://uos.edu.pk/> | Final grade: CGPA 3.90 / 4.00

PUBLICATIONS

[2026]

A Deployment Risk Score Framework for Self-Supervised Chest X-Ray Classification: Calibrated Multi-Objective Evaluation Under Annotation Scarcity and Scanner Heterogeneity

- Conceived and led a comparative study of six models (contrastive and reconstruction-based self-supervised pretraining plus four supervised baselines) on a 17,988-image, six-class chest X-ray benchmark, evaluated across 10%, 20%, and 100% annotation budgets.
- Achieved macro F1 = 0.952 with SimCLR + ResNet-50, statistically comparable to the fully supervised ceiling while retaining 88.8% of full-label performance at only 10% annotation; MAE + ViT-S/16 showed the strongest robustness to synthetic perturbations and the lowest overconfidence under calibration testing.
- Introduced a Deployment Risk Score (DRS) framework translating these accuracy, robustness, and calibration trade-offs into scenario-specific model-selection guidance for clinical deployment.

Authors: Kashif Mahmood, Romana Aziz, Muhammad Ramzan, Mahwish Ilyas, Ala Saleh Alluhaidan | **Journal Name:** Scientific Reports, In Revision | **Publisher:** Nature Portfolio

[2026]

Deep Learning-Based Disease Detection in Peanut Cultivars Utilizing Transfer Learning

- Evaluated EfficientNet-B0, EfficientNet-B4, and ConvNeXt-Base as initial candidates, none of which reached state-of-the-art performance, then developed a ResNet-50 transfer-learning approach combining data augmentation, a OneCycleLR schedule, and weighted loss to handle class imbalance.
- Achieved 97.28% accuracy on a 1,720-image benchmark of healthy and diseased groundnut leaf images, surpassing the previous best published result of 96.51% on the same dataset.
- Established a new baseline for automated groundnut leaf disease detection, demonstrating the feasibility of AI-based tools for sustainable agricultural practice.

Authors: Kashif Mahmood, Qaisar Abbas, Ume Habiba, Mahwish Ilyas, Turki Alghamdi, Hani Almoamari, Muhammad Ramzan | **Journal Name:** Scientific Reports, Under Review | **Publisher:** Nature Portfolio

[2026]

A Systematic Review of Visual Computing Technologies for Intelligent Peanut Agriculture

- Reviewed deep learning, machine learning, and hybrid approaches across four core tasks in peanut agriculture: variety classification, maturity assessment, disease detection, and pest recognition.
- Compared benchmark datasets and evaluation metrics across the literature, identifying dataset scale, model complexity, and field adaptability as the primary open challenges.
- Proposed future research directions, including lightweight visual models, multi-task integrated systems, and soil-penetrating sensing technologies.

Authors: Ume Habiba, Nabeel Abbas, Kashif Mahmood, Mahwish Ilyas, Muhammad Summair Raza, Muhammad Ramzan | **Journal Name:** USJICT, Under Review | **Publisher:** University of Sindh, Jamshoro

RESEARCH OUTPUTS

PakGroundnut-4V: A Multi-Condition Image Dataset for Pakistani Groundnut (*Arachis hypogaea* L.) Variety Classification

- Created a dataset of 8,944 images across four Pakistani peanut varieties (verified by the Barani Research Institute, Chakwal), captured across five backgrounds, four lighting conditions, and five quantity levels.
- Collected morphometric measurements (length, width, weight, seeds per pod, shell thickness) using a digital caliper and weighing scale, and published complete reproducibility documentation (CITATION.cff, data card, license, README).
- Publicly available on Zenodo: <https://doi.org/10.5281/zenodo.22014996>.

Image Dataset Cropper – Peanut Dataset Preparation Tool

- Built a full-stack local web application (Python 3.11, FastAPI, Pillow, React 18, Vite) to batch-crop composite dataset photographs into individual ML-ready images, solving the bottleneck behind preparing PakGroundnut-4V.
- Designed a custom HTML5 Canvas multi-region crop editor supporting unlimited simultaneous crop regions with drag, resize, and one-click batch-apply across an entire folder.
- Engineered for memory efficiency at scale: chunked uploads, generator-based streaming ZIP export, and disk-cached thumbnails, enabling processing of multi-gigabyte datasets with no size limit.
- Publicly available on GitHub: <https://github.com/Kashif-Mahmood007/image-dataset-cropper>

PROJECTS

PeanutInfo – Smart Agriculture System (Final Year Project)

- Built a three-module deep learning system for peanut variety classification, pest identification, and disease detection, combining a custom-collected classification dataset with two public datasets for the pest and disease modules.
- Evaluated ResNet-50 and EfficientNet-B0/B4 before achieving state-of-the-art performance with ConvNeXt (Tiny, Small, Base variants), exceeding 98% accuracy across modules.
- Deployed a full-stack web platform (HTML/CSS/Bootstrap/JavaScript frontend, PHP/MySQL backend, Flask API) for real-time image-based predictions.

PipeSense – Pipeline Condition Prediction System

- Built a regression model predicting gas and water pipeline condition from physical and environmental features, including size, diameter, material, strength, pressure, material loss, and temperature.
- Applied a Yeo-Johnson transformation, feature scaling, and ordinal and one-hot encoding within a scikit-learn Pipeline and ColumnTransformer for reproducible preprocessing.
- Achieved 97.4% accuracy, validated via 5-fold cross-validation.

Automotive Listings Scraper

- Built a Python and BeautifulSoup web scraper extracting 24,000+ structured vehicle listings (price, mileage, fuel type, transmission, engine capacity) from a local automotive platform.
- Implemented pagination handling and modular, maintainable code to scrape reliably at scale across multiple pages.
- Cleaned and exported the dataset with pandas into a CSV for market analysis and ML applications.

TECHNICAL AND SOFT SKILLS

Programming & Core Technologies:

C++, JavaScript, PHP, Python, MySQL, Pandas, NumPy, Scikit-learn, PyTorch, Matplotlib, Seaborn, BeautifulSoup.

Tools & Platforms

Jupyter Notebook, Power BI, MySQL Workbench, Git & GitHub, JIRA, Visual Studio Code, Microsoft Project, Hopsworks, GitHub Actions, CI/CD Pipeline, Streamlit

Soft Skills

Leadership, Communication, Critical Thinking, Problem Solving, Story Telling

LEADERSHIP AND COMMUNITY ENGAGEMENT

Vice President – Technical Wing, SE Spectrum

- Coordinated technical events and workshops within the Department, and delivered seminars including *"Bridging the Gap Between Data and Business Goals," "Data-Driven Decisions for Startups,"* and *"Programmatic Thinking."*
- Mentored peers in technical domains, encouraging participation in research and practical projects.

Head – Data Science Group, Techive Society (University of Sargodha)

- Led the faculty's Data Science group, planning seminars and learning sessions focused on AI, ML, and data analytics.
- Supported more than 100 junior students through workshops, technical sessions, and informal mentorship in AI, ML, and data analytics tools.

HONOURS AND CERTIFICATIONS

Awards & Recognition

- 3rd Position, Annual University Sports Gala – Badminton (representing the Department of Software Engineering)
- Champion, Inter-Departmental Badminton (Solo & Doubles) — qualifying selection event for the Annual Sports Gala
- Certificates of Appreciation for seminars delivered on data-business alignment, startup analytics, and problem-solving

Certifications

- Completed 6 courses in data science, Python, and analytics (Udemy, Coursera)

[Certificates available upon request](#)